

BLM2041 Signals and Systems

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BLM2041 Signals and Systems

Spectrum Representation

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Problem Solving Skills

- Math Formula
 - Sum of Cosines
 - Amp, Freq, Phase
- Recorded Signals
 - Speech
 - Music
 - No simple formula
- Plot & Sketches
 - $S(t)$ versus t
 - Spectrum
- MATLAB
 - Numerical
 - Computation
 - Plotting list of numbers



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LECTURE OBJECTIVES

- Sinusoids with **DIFFERENT** Frequencies
 - **SYNTHESIZE** by Adding Sinusoids

$$x(t) = \sum_{k=1}^N A_k \cos(2\pi f_k t + \varphi_k)$$

- **SPECTRUM** Representation
 - Graphical Form shows **DIFFERENT** Freqs



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LECTURE OBJECTIVES

- Signals with **HARMONIC** Frequencies
 - Add Sinusoids with $f_k = kf_0$

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi k f_0 t + \varphi_k)$$

FREQUENCY can change vs. TIME

Chirps:

$$x(t) = \cos(\alpha t^2)$$

Introduce Spectrogram Visualization (`specgram.m`)
(`plotspec.m`)

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LECTURE OBJECTIVES

- Work with the Fourier Series Integral

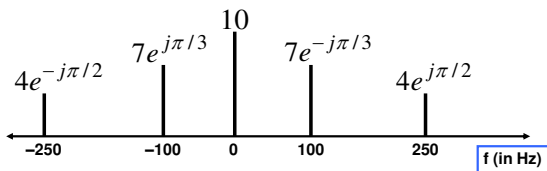
$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi k/T_0)t} dt$$

- **ANALYSIS** via Fourier Series
 - For **PERIODIC** signals: $x(t+T_0) = x(t)$
- **SPECTRUM** from Fourier Series
 - a_k is Complex Amplitude for k -th Harmonic

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FREQUENCY DIAGRAM

- Plot Complex Amplitude vs. Freq



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Another FREQ. Diagram

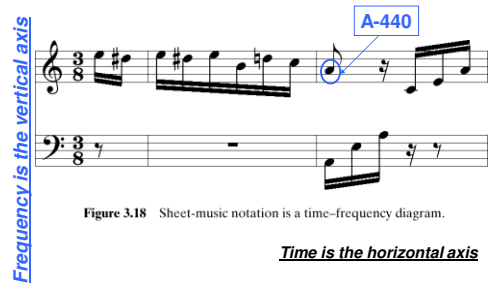


Figure 3.18 Sheet-music notation is a time-frequency diagram.

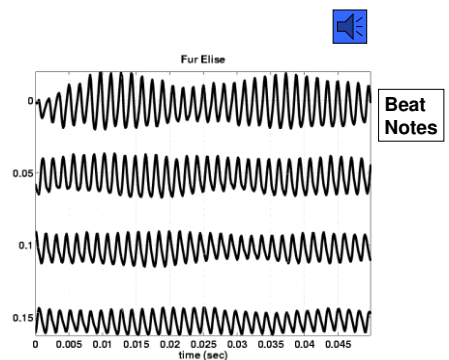
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MOTIVATION

- Synthesize **Complicated** Signals
 - Musical Notes**
 - Piano uses 3 strings for many notes
 - Chords: play several notes simultaneously
 - Human Speech**
 - Vowels have dominant frequencies
 - Application: computer generated speech
 - Can **all** signals be generated this way?
 - Sum of sinusoids?

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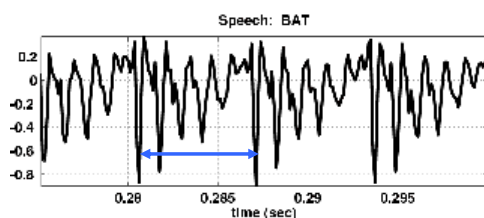
Fur Elise WAVEFORM



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Speech Signal: BAT

- Nearly **Periodic** in Vowel Region
 - Period is (Approximately) $T = 0.0065$ sec



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Euler's Formula Reversed

- Solve for **cosine** (or sine)

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$

$$e^{-j\omega t} = \cos(-\omega t) + j\sin(-\omega t)$$

$$e^{-j\omega t} = \cos(\omega t) - j\sin(\omega t)$$

$$e^{j\omega t} + e^{-j\omega t} = 2\cos(\omega t)$$

$$\cos(\omega t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t})$$

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INVERSE Euler's Formula

- Solve for **cosine** (or sine)

$$\cos(\omega t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t})$$

$$\sin(\omega t) = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t})$$

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SPECTRUM Interpretation

- Cosine = sum of 2 complex exponentials:

$$A \cos(7t) = \frac{A}{2} e^{j7t} + \frac{A}{2} e^{-j7t}$$

One has a positive frequency
The other has **negative** freq.
Amplitude of each is half as big

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NEGATIVE FREQUENCY

- Is negative frequency real?
- Doppler Radar provides an example
 - Police radar measures speed by using the Doppler shift principle
 - Let's assume 400Hz $\leftrightarrow$ 60 mph
 - +400Hz means towards the radar
 - 400Hz means away (opposite direction)
 - Think of a train whistle

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SPECTRUM of SINE

- Sine = sum of 2 complex exponentials:

$$\begin{aligned} A \sin(7t) &= \frac{A}{2j} e^{j7t} - \frac{A}{2j} e^{-j7t} \\ &= \frac{1}{2} A e^{-j0.5\pi} e^{j7t} + \frac{1}{2} A e^{j0.5\pi} e^{-j7t} \end{aligned}$$

$$\frac{-1}{j} = j = e^{j0.5\pi}$$

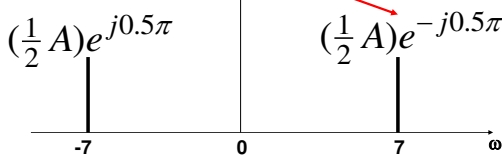
- Positive freq. has phase = -0.5π
- Negative freq. has phase = $+0.5\pi$

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GRAPHICAL SPECTRUM

EXAMPLE of SINE

$$A \sin(7t) = \frac{1}{2} A e^{-j0.5\pi} e^{j7t} + \frac{1}{2} A e^{j0.5\pi} e^{-j7t}$$

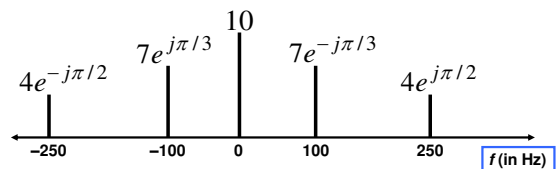


AMPLITUDE, PHASE & FREQUENCY are shown

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SPECTRUM ---> SINUSOID

- Add the spectrum components:



What is the formula for the signal $x(t)$?

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Gather (A,ω,φ) information

Frequencies:	Amplitude & Phase
-250 Hz	-4 $-\pi/2$
-100 Hz	-7 $+\pi/3$
0 Hz	-10 0
100 Hz	-7 $-\pi/3$
250 Hz	-4 $+\pi/2$

Note the **conjugate phase**

DC is another name for zero-freq component

DC component always has $\phi=0$ or π (for real $x(t)$)

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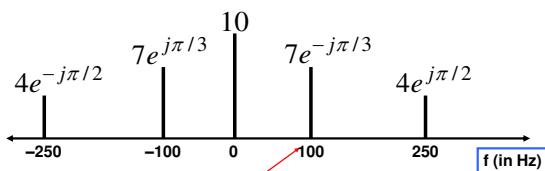
Add Spectrum Components-1

Frequencies:	Amplitude & Phase
-250 Hz	-4 $-\pi/2$
-100 Hz	-7 $+\pi/3$
0 Hz	-10 0
100 Hz	-7 $-\pi/3$
250 Hz	-4 $+\pi/2$

$$x(t) = 10 + 7e^{-j\pi/3}e^{j2\pi(100)t} + 7e^{j\pi/3}e^{-j2\pi(100)t} + 4e^{j\pi/2}e^{j2\pi(250)t} + 4e^{-j\pi/2}e^{-j2\pi(250)t}$$

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Add Spectrum Components-2



$$x(t) = 10 + 7e^{-j\pi/3}e^{j2\pi(100)t} + 7e^{j\pi/3}e^{-j2\pi(100)t} + 4e^{j\pi/2}e^{j2\pi(250)t} + 4e^{-j\pi/2}e^{-j2\pi(250)t}$$

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Simplify Components

$$x(t) = 10 + 7e^{-j\pi/3}e^{j2\pi(100)t} + 7e^{j\pi/3}e^{-j2\pi(100)t} + 4e^{j\pi/2}e^{j2\pi(250)t} + 4e^{-j\pi/2}e^{-j2\pi(250)t}$$

Use Euler's Formula to get **REAL** sinusoids:

$$A\cos(\omega t + \varphi) = \frac{1}{2}Ae^{-j\varphi}e^{j\omega t} + \frac{1}{2}Ae^{-j\varphi}e^{-j\omega t}$$

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FINAL ANSWER

$$x(t) = 10 + 14\cos(2\pi(100)t - \pi/3) + 8\cos(2\pi(250)t + \pi/2)$$

So, we get the general form:

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi f_k t + \varphi_k)$$

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Summary: GENERAL FORM

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi f_k t + \varphi_k)$$

$$x(t) = X_0 + \sum_{k=1}^N \Re\{X_k e^{j2\pi f_k t}\} \quad \begin{matrix} X_k = A_k e^{j\varphi_k} \\ \text{Frequency} = f_k \end{matrix}$$

$$\Re\{z\} = \frac{1}{2}z + \frac{1}{2}z^*$$

$$x(t) = X_0 + \sum_{k=1}^N \left\{ \frac{1}{2}X_k e^{j2\pi f_k t} + \frac{1}{2}X_k^* e^{-j2\pi f_k t} \right\}$$

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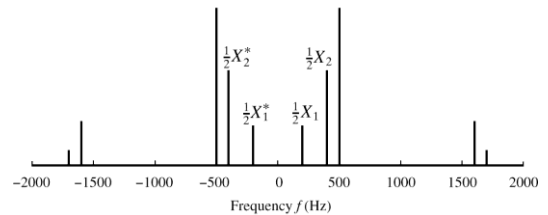
Example: Synthetic Vowel

- Sum of 5 Frequency Components
 - Complex amplitudes for harmonic signal that approximates the vowel sound «ah»

f_k (Hz)	X_k	Mag	Phase (rad)
200	$(771 + j12202)$	12,226	1.508
400	$(-8865 + j28048)$	29,416	1.876
500	$(48001 - j8995)$	48,836	-0.185
1600	$(1657 - j13520)$	13,621	-1.449
1700	$4723 + j0$	4723	0

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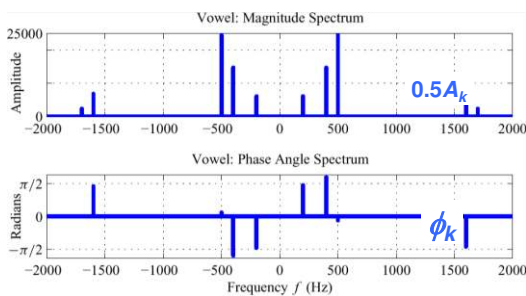
SPECTRUM of VOWEL



- Note: Spectrum has $0.5X_k$ (except X_{DC})
- Conjugates in negative frequency

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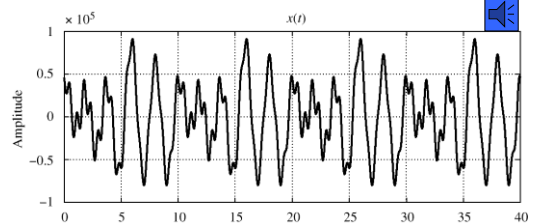
SPECTRUM of VOWEL (Polar Format)



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Vowel Waveform (sum of all 5 components)

- Sum of all of the signals in the previous slides



- Note that the period is 10 ms, which equals $1/f_0$

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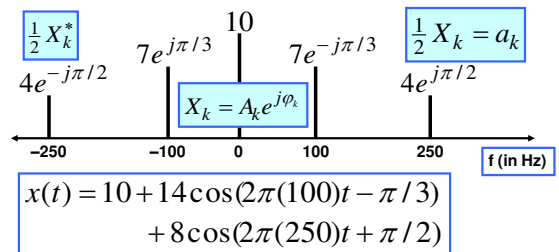
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Periodic Signals, Harmonics & Time-Varying Sinusoids

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SPECTRUM DIAGRAM

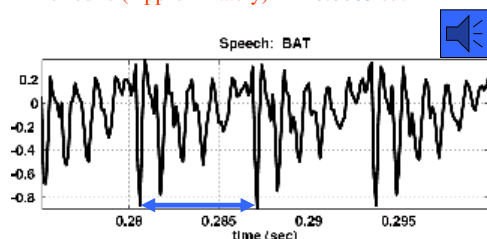
- Recall Complex Amplitude vs. Freq



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SPECTRUM for PERIODIC ?

- Nearly **Periodic** in the Vowel Region
 - Period is (Approximately) $T = 0.0065$ sec



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PERIODIC SIGNALS

- Repeat every T secs

– Definition

– Example:

$$x(t) = x(t+T)$$

$$x(t) = \cos^2(3t)$$

$$T = ?$$

– Speech can be “quasi-periodic”

$$T = \frac{2\pi}{3}$$

$$T = \frac{\pi}{3}$$

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Period of Complex Exponential

$$x(t) = e^{j\omega t}$$

$$x(t+T) = x(t) ?$$

Definition: Period is T

$$e^{j\omega(t+T)} = e^{j\omega t}$$

$$\Rightarrow e^{j\omega T} = 1 \Rightarrow \omega T = 2\pi k$$

$$e^{j2\pi k} = 1$$

$$\omega = \frac{2\pi k}{T} = \left(\frac{2\pi}{T}\right)k = \omega_0 k$$

$k = \text{integer}$

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Harmonic Signal Spectrum

Periodic signal can only have: $f_k = k f_0$

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi k f_0 t + \varphi_k)$$

$$f_0 = \frac{1}{T}$$

$$X_k = A_k e^{j\varphi_k}$$

$$x(t) = X_0 + \sum_{k=1}^N \left\{ \frac{1}{2} X_k e^{j2\pi k f_0 t} + \frac{1}{2} X_k^* e^{-j2\pi k f_0 t} \right\}$$

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Define FUNDAMENTAL FREQ

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi k f_0 t + \varphi_k)$$

$$f_k = k f_0 \quad (\omega_0 = 2\pi f_0)$$

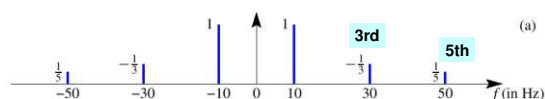
$$f_0 = \frac{1}{T_0}$$

f_0 = fundamental Frequency (largest)

T_0 = fundamental Period (shortest)

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Harmonic Signal (3 Freqs)



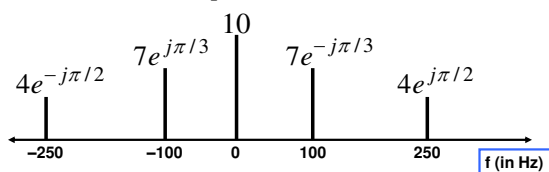
What is the fundamental frequency?

10 Hz

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POP QUIZ: FUNDAMENTAL

- Here's another spectrum:



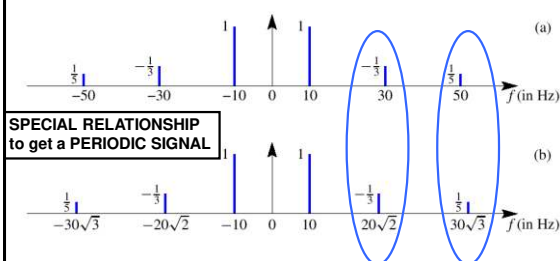
What is the fundamental frequency?

100 Hz ?

50 Hz ?

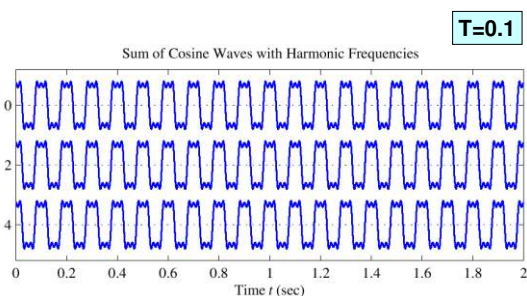
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IRRATIONAL SPECTRUM



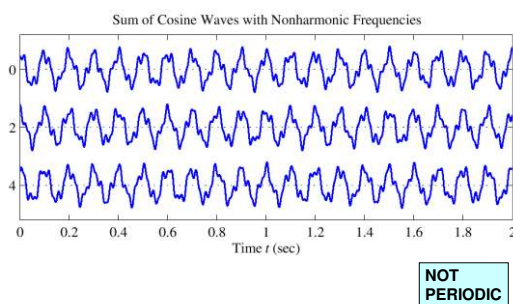
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Harmonic Signal (3 Freqs)



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NON-Harmonic Signal



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FREQUENCY ANALYSIS

- Now, a much HARDER problem**
- Given a recording of a song, have the computer write the music



- Can a machine extract frequencies?
 - Yes, if we COMPUTE the spectrum for $x(t)$
 - During short intervals

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Time-Varying FREQUENCIES Diagram

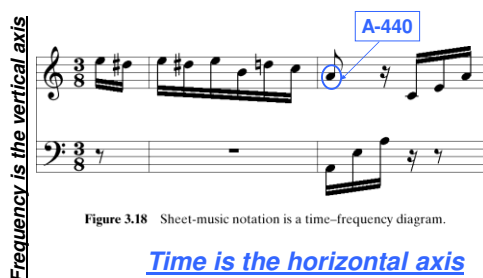
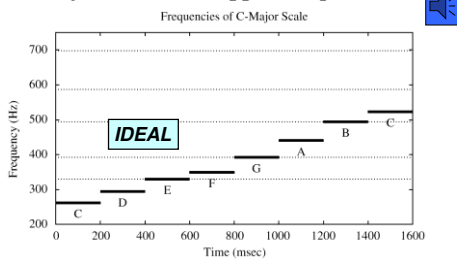


Figure 3.18 Sheet-music notation is a time-frequency diagram.

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SIMPLE TEST SIGNAL

- C-major SCALE: stepped frequencies



– Frequency is constant for each note

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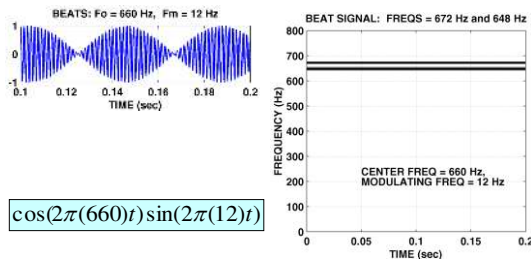
SPECTROGRAM

- SPECTROGRAM Tool
 - MATLAB function is `spectgram.m`
 - SP-First has `plotspec.m` & `spectgr.m`
- ANALYSIS program
 - Takes $x(t)$ as input &
 - Produces spectrum values X_k
 - Breaks $x(t)$ into SHORT TIME SEGMENTS
 - Then uses the FFT (Fast Fourier Transform)

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SPECTROGRAM EXAMPLE

- Two Constant Frequencies: Beats



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AM Radio Signal

- Same as BEAT Notes

$$\cos(2\pi(660)t) \sin(2\pi(12)t)$$

$$\frac{1}{2} \left(e^{j2\pi(660)t} + e^{-j2\pi(660)t} \right) \frac{1}{2j} \left(e^{j2\pi(12)t} - e^{-j2\pi(12)t} \right)$$

$$\frac{1}{4j} \left(e^{j2\pi(672)t} - e^{-j2\pi(672)t} - e^{j2\pi(648)t} + e^{-j2\pi(648)t} \right)$$

$$\frac{1}{2} \cos(2\pi(672)t - \frac{\pi}{2}) + \frac{1}{2} \cos(2\pi(648)t + \frac{\pi}{2})$$

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SPECTRUM of AM (Beat)

- 4 complex exponentials in AM:



What is the fundamental frequency?

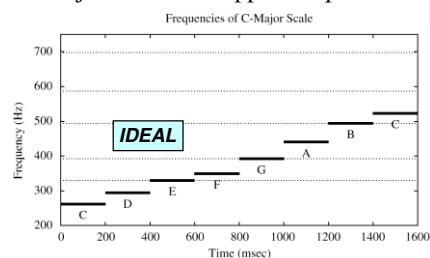
648 Hz ?

24 Hz ?

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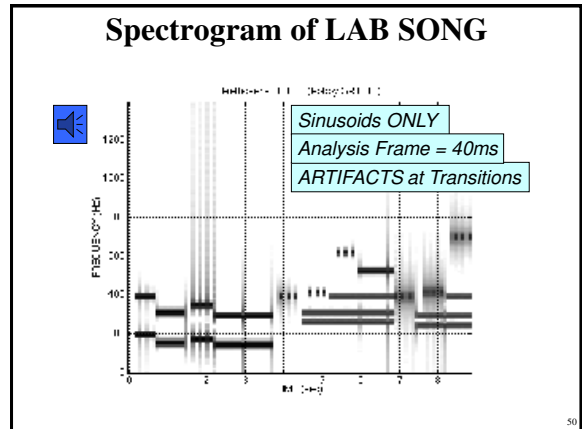
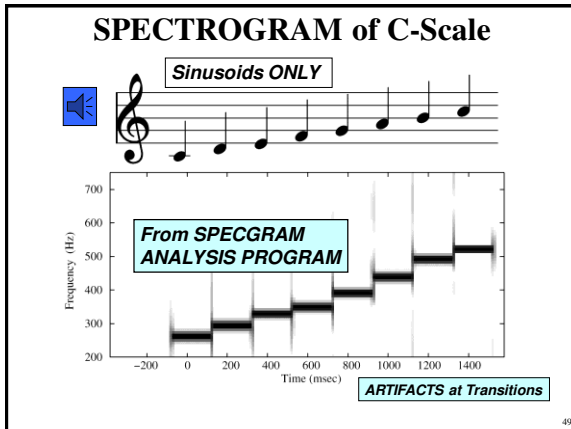
STEPPED FREQUENCIES

- C-major SCALE: stepped frequencies



– Frequency is constant for each note

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Time-Varying Frequency

- Frequency can change **vs. time**
 - Continuously, not stepped
- FREQUENCY MODULATION (FM)**

$$x(t) = \cos(2\pi f_c t + v(t))$$

VOICE
- CHIRP SIGNALS
 - Linear Frequency Modulation (LFM)

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New Signal: Linear FM

- Called **Chirp** Signals (LFM)
 - Quadratic phase

$$x(t) = A \cos(\alpha t^2 + 2\pi f_0 t + \varphi)$$

QUADRATIC

- Freq will change **LINEARLY** vs. time
 - Example of Frequency Modulation (FM)
 - Define “instantaneous frequency”

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INSTANTANEOUS FREQ

- Definition

$$x(t) = A \cos(\psi(t))$$

$$\Rightarrow \omega_i(t) = \frac{d}{dt} \psi(t)$$

Derivative of the “Angle”
- For Sinusoid:

$$x(t) = A \cos(2\pi f_0 t + \varphi)$$

$$\psi(t) = 2\pi f_0 t + \varphi$$

Makes sense

$$\Rightarrow \omega_i(t) = \frac{d}{dt} \psi(t) = 2\pi f_0$$

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INSTANTANEOUS FREQ of the Chirp

- Chirp** Signals have Quadratic phase
- Freq will change **LINEARLY** vs. time

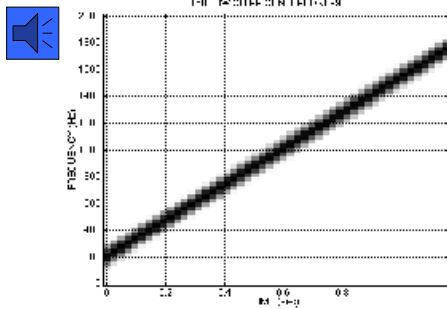
$$x(t) = A \cos(\alpha t^2 + \beta t + \varphi)$$

$$\Rightarrow \psi(t) = \alpha t^2 + \beta t + \varphi$$

$$\Rightarrow \omega_i(t) = \frac{d}{dt} \psi(t) = 2\alpha t + \beta$$

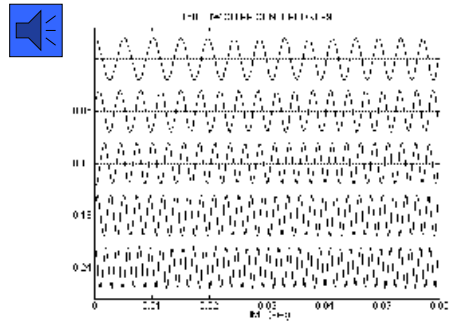
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CHIRP SPECTROGRAM



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CHIRP WAVEFORM



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OTHER CHIRPS

$\psi(t)$ can be anything:

$$x(t) = A \cos(\alpha \cos(\beta t) + \varphi)$$

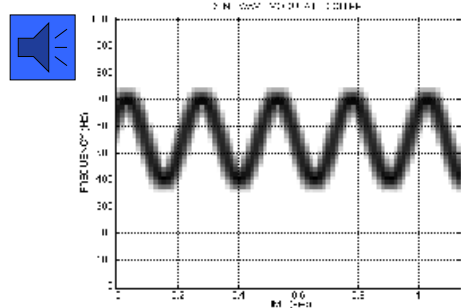
$$\Rightarrow \omega_i(t) = \frac{d}{dt} \psi(t) = -\alpha \beta \sin(\beta t)$$

$\psi(t)$ could be speech or music:

– FM radio broadcast

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SINE-WAVE FREQUENCY MODULATION (FM)



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Fourier Series Coefficients

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HISTORY

- Jean Baptiste Joseph Fourier (1768-1830)



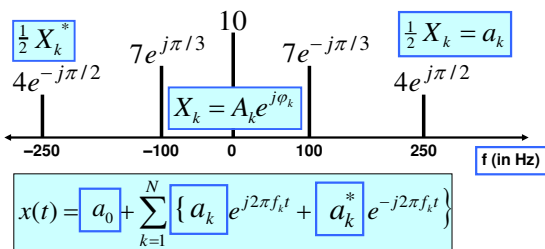
- Napoleonic era
- Studied the mathematical theory of heat conduction
- Established the partial differential equation governing heat diffusion and solved it by using infinite series of trigonometric functions.

- <http://www-groups.dcs.st-and.ac.uk/~history/Mathematicians/Fourier.html>

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SPECTRUM DIAGRAM

- Recall Complex Amplitude vs. Freq



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Harmonic Signal

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi k f_0 t}$$

PERIOD/FREQUENCY of COMPLEX EXPONENTIAL:

$$2\pi(f_0) = \omega_0 = \frac{2\pi}{T_0} \quad \text{or} \quad T_0 = \frac{1}{f_0}$$

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Fourier Series Synthesis

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi k f_0 t}$$

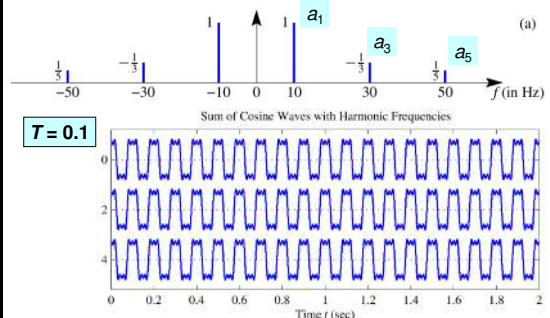
$$a_k = \frac{1}{2} X_k = \frac{1}{2} A_k e^{j\varphi_k}$$

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi k f_0 t + \varphi_k)$$

$X_k = A_k e^{j\varphi_k}$ ← COMPLEX AMPLITUDE

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Harmonic Signal (3 Freqs)



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SYNTHESIS vs. ANALYSIS

- | | |
|--|--|
| <ul style="list-style-type: none"> SYNTHESIS <ul style="list-style-type: none"> – Easy – Given (ω_k, A_k, ϕ_k) create $x(t)$ Synthesis can be HARD <ul style="list-style-type: none"> – Synthesize Speech so that it sounds good | <ul style="list-style-type: none"> ANALYSIS <ul style="list-style-type: none"> – Hard – Given $x(t)$, extract (ω_k, A_k, ϕ_k) – How many? – Need algorithm for computer |
|--|--|

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STRATEGY: $x(t) \rightarrow a_k$

- ANALYSIS**
 - Get representation from the signal
 - Works for **PERIODIC** Signals
- Fourier Series
 - Answer is: an INTEGRAL over one period

$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j\omega_0 k t} dt$$

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INTEGRAL Property of $\exp(j)$

- INTEGRATE over ONE PERIOD

$$\int_0^{T_0} e^{-j(2\pi/T_0)mt} dt = \frac{T_0}{-j2\pi m} e^{-j(2\pi/T_0)mt} \Big|_0^{T_0}$$

$$= \frac{T_0}{-j2\pi m} (e^{-j2\pi m} - 1)$$

$$\int_0^{T_0} e^{-j(2\pi/T_0)mt} dt = 0 \quad m \neq 0 \quad \omega_0 = \frac{2\pi}{T_0}$$

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ORTHOGONALITY of $\exp(j)$

- PRODUCT of $\exp(+j)$ and $\exp(-j)$

$$\frac{1}{T_0} \int_0^{T_0} e^{j(2\pi/T_0)\ell t} e^{-j(2\pi/T_0)kt} dt = \begin{cases} 0 & k \neq \ell \\ 1 & k = \ell \end{cases}$$

$$\frac{1}{T_0} \int_0^{T_0} e^{j(2\pi/T_0)(\ell-k)t} dt$$

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Isolate One FS Coefficient

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j(2\pi/T_0)kt}$$

$$\frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)\ell t} dt = \frac{1}{T_0} \int_0^{T_0} \left(\sum_{k=-\infty}^{\infty} a_k e^{j(2\pi/T_0)kt} \right) e^{-j(2\pi/T_0)\ell t} dt$$

$$\frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)\ell t} dt = \sum_{k=-\infty}^{\infty} a_k \left(\frac{1}{T_0} \int_0^{T_0} e^{j(2\pi/T_0)kt} e^{-j(2\pi/T_0)\ell t} dt \right) = a_\ell$$

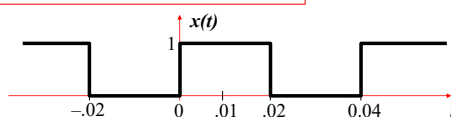
$$\Rightarrow a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)kt} dt \quad \text{Integral is zero except for } k = \ell$$

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SQUARE WAVE EXAMPLE

$$x(t) = \begin{cases} 1 & 0 \leq t < \frac{1}{2}T_0 \\ 0 & \frac{1}{2}T_0 \leq t < T_0 \end{cases}$$

for $T_0 = 0.04$ sec.



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FS for a SQUARE WAVE $\{a_k\}$

$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)kt} dt \quad (k \neq 0)$$

$$a_k = \frac{1}{.04} \int_0^{.02} 1 e^{-j(2\pi/.04)kt} dt = \frac{1}{.04(-j2\pi k/.04)} e^{-j(2\pi/.04)kt} \Big|_0^{.02}$$

$$= \frac{1}{(-j2\pi k)} (e^{-j(\pi)k} - 1) = \frac{1 - (-1)^k}{j2\pi k}$$

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DC Coefficient: a_0

$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)kt} dt \quad (k = 0)$$

$$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt = \frac{1}{T_0} (\text{Area})$$

$$a_0 = \frac{1}{.04} \int_0^{.02} 1 dt = \frac{1}{.04} (.02 - 0) = \frac{1}{2}$$

72

Fourier Coefficients a_k

- a_k is a function of k
 - Complex Amplitude for k -th Harmonic
 - This one doesn't depend on the period, T_0

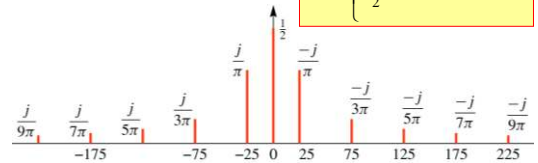
$$a_k = \frac{1 - (-1)^k}{j2\pi k} = \begin{cases} \frac{1}{j\pi k} & k = \pm 1, \pm 3, \dots \\ 0 & k = \pm 2, \pm 4, \dots \\ \frac{1}{2} & k = 0 \end{cases}$$

73

Spectrum from Fourier Series

$$\omega_0 = 2\pi / (0.04) = 2\pi(25)$$

$$a_k = \begin{cases} \frac{-j}{\pi k} & k = \pm 1, \pm 3, \dots \\ 0 & k = \pm 2, \pm 4, \dots \\ \frac{1}{2} & k = 0 \end{cases}$$



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Fourier Series Integral

- HOW do you determine a_k from $x(t)$?

$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)kt} dt$$

Fundamental Frequency $f_0 = 1/T_0$

$$a_{-k} = a_k^* \quad \text{when } x(t) \text{ is real}$$

$$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt \quad (\text{DC component})$$

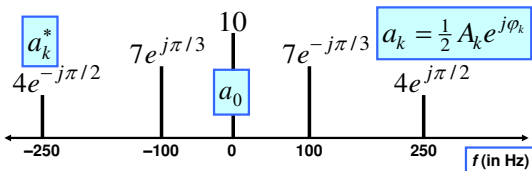
75

Fourier Series & Spectrum

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SPECTRUM DIAGRAM

- Recall Complex Amplitude vs. Freq



$$x(t) = a_0 + \sum_{k=1}^N \left\{ a_k e^{j2\pi f_k t} + a_k^* e^{-j2\pi f_k t} \right\}$$

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Harmonic Signal

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi k f_0 t}$$

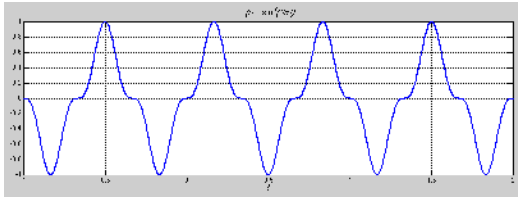
PERIOD/FREQUENCY of COMPLEX EXPONENTIAL:

$$2\pi(f_0) = \omega_0 = \frac{2\pi}{T_0} \quad \text{or} \quad T_0 = \frac{1}{f_0}$$

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Example

$$x(t) = \sin^3(3\pi t)$$



$$x(t) = \left(\frac{j}{8}\right)e^{j9\pi t} + \left(\frac{-3j}{8}\right)e^{j3\pi t} + \left(\frac{3j}{8}\right)e^{-j3\pi t} + \left(\frac{-j}{8}\right)e^{-j9\pi t}$$

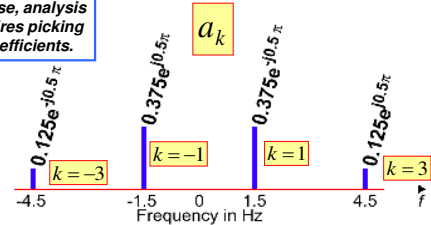
79

Example

$$x(t) = \sin^3(3\pi t)$$

$$x(t) = \left(\frac{j}{8}\right)e^{j9\pi t} + \left(\frac{-3j}{8}\right)e^{j3\pi t} + \left(\frac{3j}{8}\right)e^{-j3\pi t} + \left(\frac{-j}{8}\right)e^{-j9\pi t}$$

In this case, analysis just requires picking off the coefficients.



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STRATEGY: $x(t) \rightarrow a_k$

• ANALYSIS

- Get representation from the signal
- Works for **PERIODIC** Signals
- Fourier Series
- Answer is: an **INTEGRAL** over one period

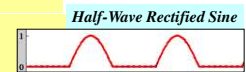
$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j\omega_0 k t} dt$$

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FS: Rectified Sine Wave $\{a_k\}$

$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)kt} dt \quad (k \neq \pm 1)$$

$$\begin{aligned} a_k &= \frac{1}{T_0} \int_0^{T_0/2} \sin\left(\frac{2\pi}{T_0}t\right) e^{-j(2\pi/T_0)kt} dt \\ &= \frac{1}{T_0} \int_0^{T_0/2} \frac{e^{j(2\pi/T_0)t} - e^{-j(2\pi/T_0)t}}{2j} e^{-j(2\pi/T_0)kt} dt \\ &= \frac{1}{j2T_0} \int_0^{T_0/2} e^{-j(2\pi/T_0)(k-1)t} dt - \frac{1}{j2T_0} \int_0^{T_0/2} e^{-j(2\pi/T_0)(k+1)t} dt \\ &= \frac{e^{-j(2\pi/T_0)(k-1)T_0/2}}{j2T_0(-j(2\pi/T_0)(k-1))} \Big|_0^{T_0/2} - \frac{e^{-j(2\pi/T_0)(k+1)T_0/2}}{j2T_0(-j(2\pi/T_0)(k+1))} \Big|_0^{T_0/2} \end{aligned}$$



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FS: Rectified Sine Wave $\{a_k\}$

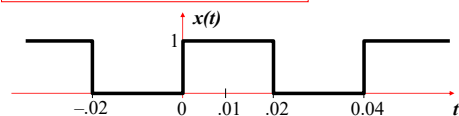
$$\begin{aligned} a_k &= \frac{e^{-j(2\pi/T_0)(k-1)T_0/2}}{j2T_0(-j(2\pi/T_0)(k-1))} \Big|_0^{T_0/2} - \frac{e^{-j(2\pi/T_0)(k+1)T_0/2}}{j2T_0(-j(2\pi/T_0)(k+1))} \Big|_0^{T_0/2} \\ &= \frac{1}{4\pi(k-1)} \left(e^{-j(2\pi/T_0)(k-1)T_0/2} - 1 \right) - \frac{1}{4\pi(k+1)} \left(e^{-j(2\pi/T_0)(k+1)T_0/2} - 1 \right) \\ &= \frac{1}{4\pi(k-1)} \left(e^{-j\pi(k-1)} - 1 \right) - \frac{1}{4\pi(k+1)} \left(e^{-j\pi(k+1)} - 1 \right) \\ &= \frac{(k+1-(k-1))}{4\pi(k^2-1)} \left((-1)^k - 1 \right) = \begin{cases} 0 & k \text{ odd} \\ \pm \frac{1}{j4} & k = \pm 1 \\ \frac{-1}{\pi(k^2-1)} & k \text{ even} \end{cases} \end{aligned}$$

83

SQUARE WAVE EXAMPLE

$$x(t) = \begin{cases} 1 & 0 \leq t < \frac{1}{2}T_0 \\ 0 & \frac{1}{2}T_0 \leq t < T_0 \end{cases}$$

for $T_0 = 0.04$ sec.



84

Fourier Coefficients a_k

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 - Complex Amplitude for k -th Harmonic
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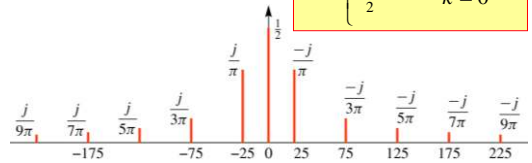
$$a_k = \frac{1 - (-1)^k}{j2\pi k} = \begin{cases} \frac{1}{j\pi k} & k = \pm 1, \pm 3, \dots \\ 0 & k = \pm 2, \pm 4, \dots \\ \frac{1}{2} & k = 0 \end{cases}$$

87

Spectrum from Fourier Series

$$\omega_0 = 2\pi / (0.04) = 2\pi(25)$$

$$a_k = \begin{cases} \frac{-j}{\pi k} & k = \pm 1, \pm 3, \dots \\ 0 & k = \pm 2, \pm 4, \dots \\ \frac{1}{2} & k = 0 \end{cases}$$



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Fourier Series Synthesis

- HOW do you **APPROXIMATE** $x(t)$?

$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(2\pi/T_0)kt} dt$$

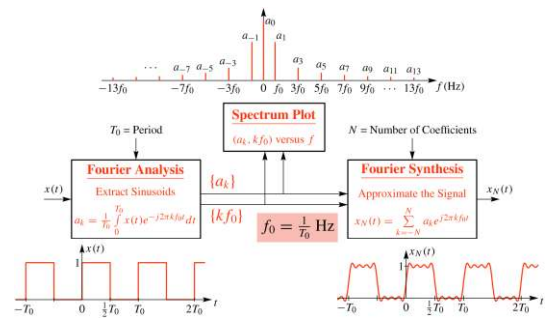
- Use FINITE number of coefficients

$$x(t) = \sum_{k=-N}^N a_k e^{j2\pi k f_0 t}$$

$a_{-k} = a_k^*$ when $x(t)$ is real

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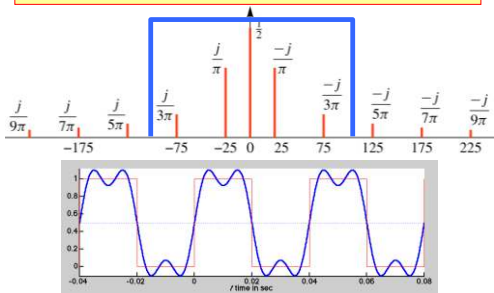
Fourier Series Synthesis



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Synthesis: 1st & 3rd Harmonics

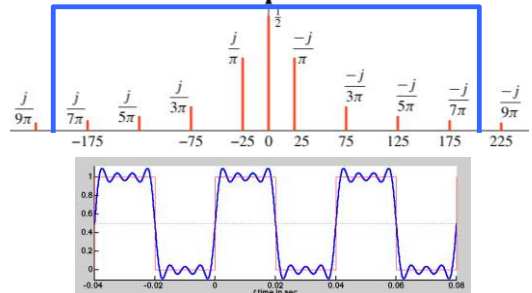
$$y(t) = \frac{1}{2} + \frac{2}{\pi} \cos(2\pi(25)t - \frac{\pi}{2}) + \frac{2}{3\pi} \cos(2\pi(75)t - \frac{\pi}{2})$$



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Synthesis: up to 7th Harmonic

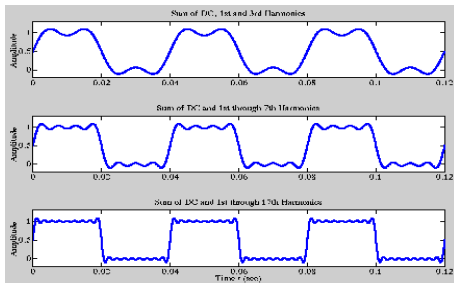
$$y(t) = \frac{1}{2} + \frac{2}{\pi} \cos(50\pi t - \frac{\pi}{2}) + \frac{2}{3\pi} \sin(150\pi t) + \frac{2}{5\pi} \sin(250\pi t) + \frac{2}{7\pi} \sin(350\pi t)$$



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Fourier Synthesis

$$x_N(t) = \frac{1}{2} + \frac{2}{\pi} \sin(\omega_0 t) + \frac{2}{3\pi} \sin(3\omega_0 t) + \dots$$



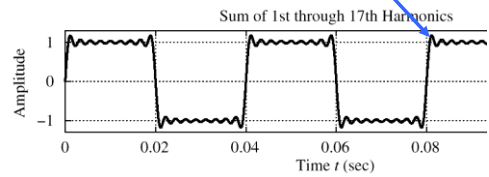
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Gibbs' Phenomenon

- Convergence at **DISCONTINUITY** of $x(t)$

– There is always an **overshoot**

– **9%** for the **Square Wave** case



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